WiDS Datathon 2026

# Wildfire Evacuation Equity

## AUDITING WILDFIRE EVACUATION IN CALIFORNIA

TRACK 1: ACCELERATING EQUITABLE EVACUATION

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Women in Data Science Worldwide

# The gap between a fire being seen and the order to leave

## The Question

“How long does it take from a fire being seen to an evacuation order, and does that gap increase for vulnerable communities?”

## ~ 50 Mins

- median time from first sighting to evacuation order

# Datasets & Schema

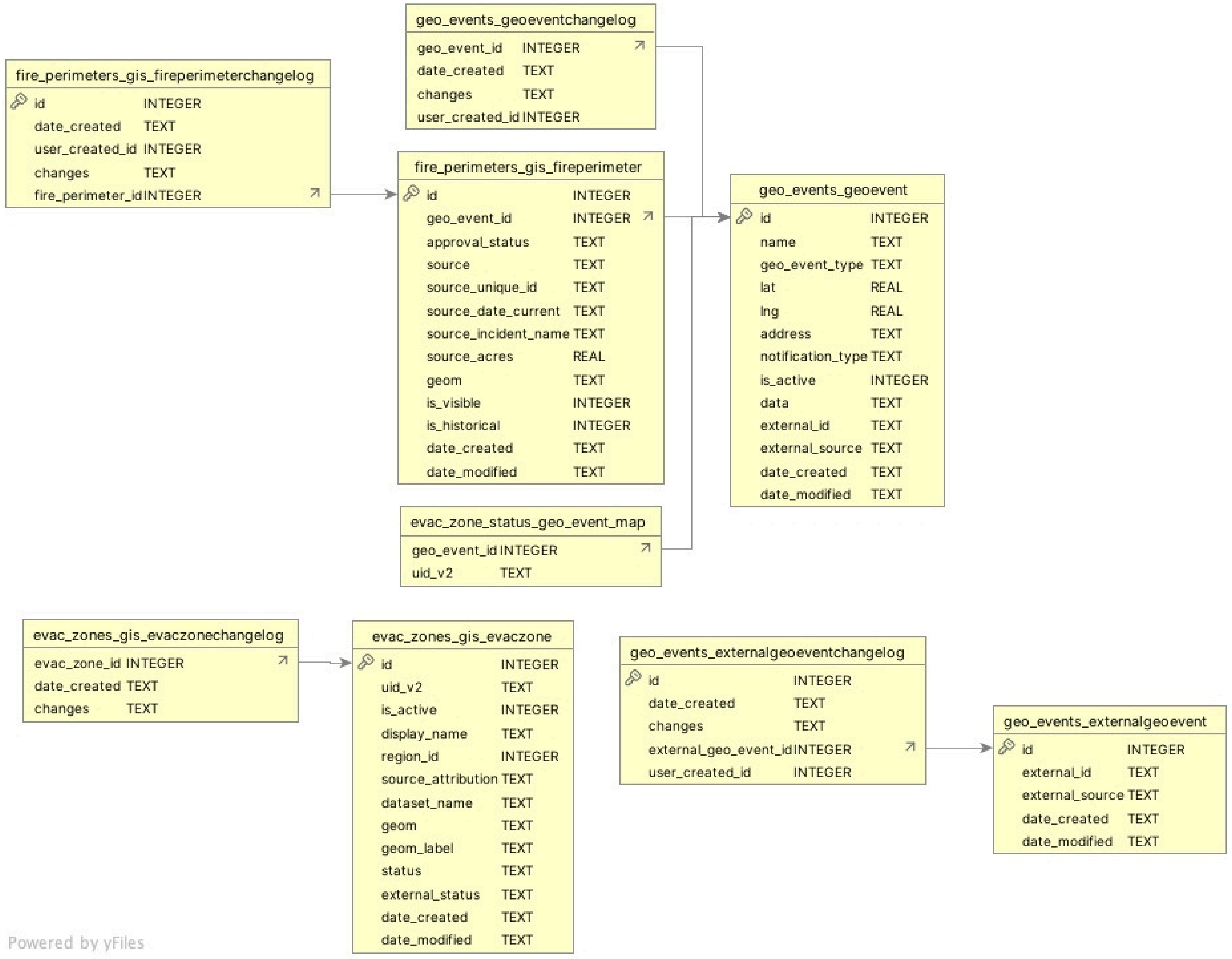


## WatchDuty data - 9 datasets

- geo_events: fire records (name, location, time, etc.)
- geo_events_changelog: reporter updates used to calculate alert lag
- evac_zones + changelog: evacuation zone status over time
- fire_perimeters + changelog: fire boundary changes over time
- external_geo_event_changelog: external source updates
- evac_zone_status_geo_event_map: links zones to fire events

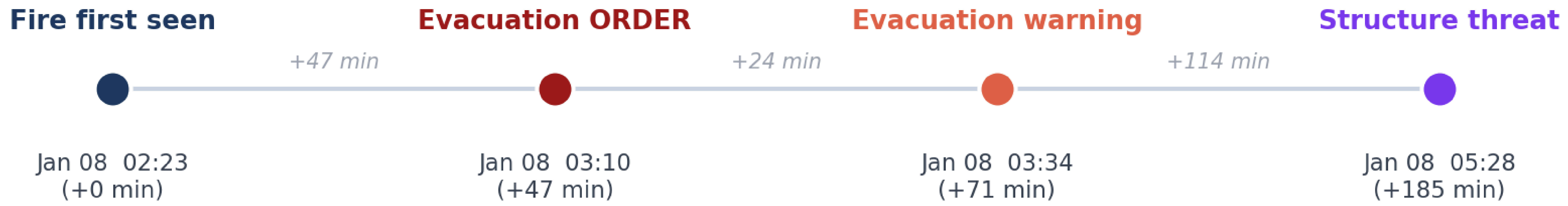
## Census data - CDC SVI 2022

- 9,109 California census tracts
- 6 indicators used to calculate the vulnerability scores of each county: no internet, no vehicle, aged 65+, limited English, disability, below poverty (150%)



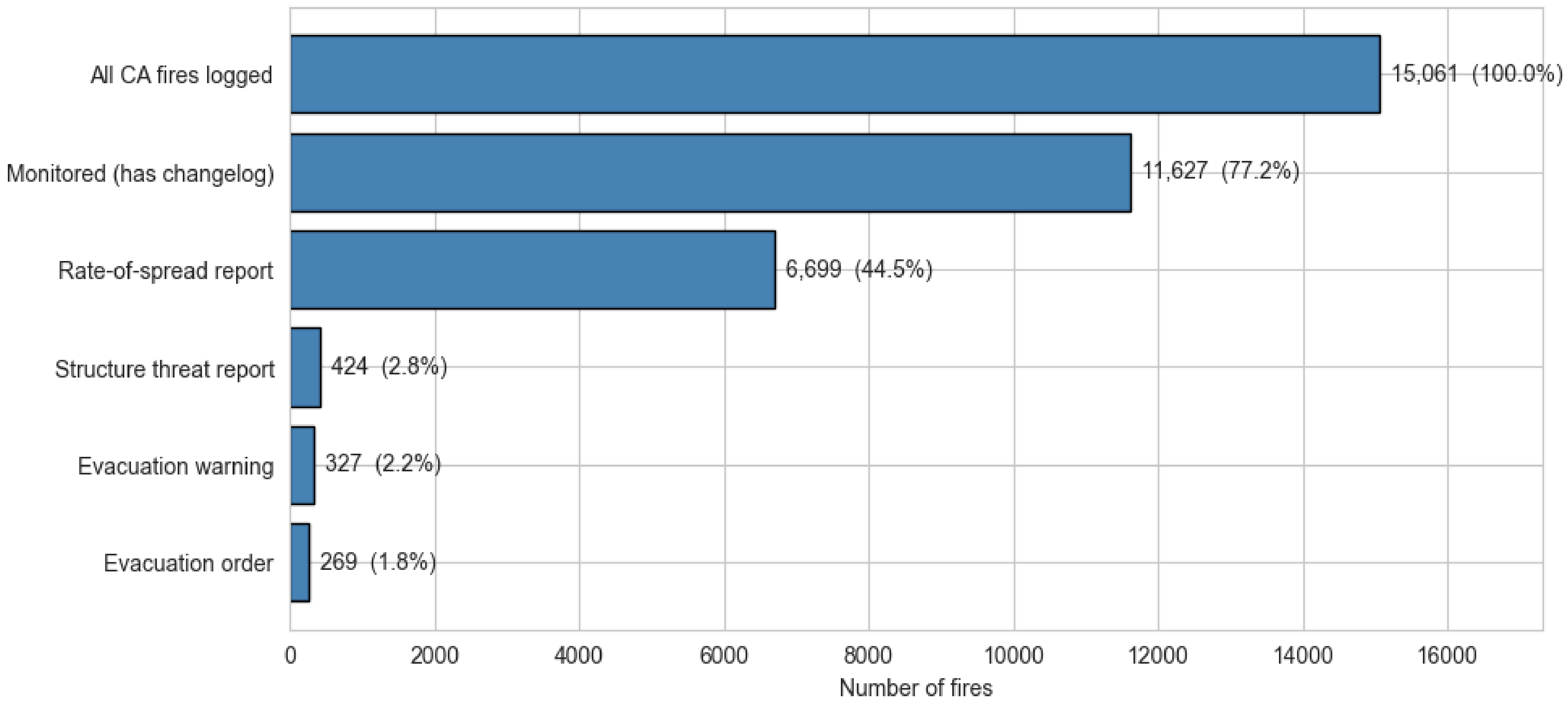
# **2025 Los Angeles Fire:** Eaton Fire Case






Evacuation orders went out 47 minutes after the fire was first recorded.

# Most fires are never escalated to evacuation order

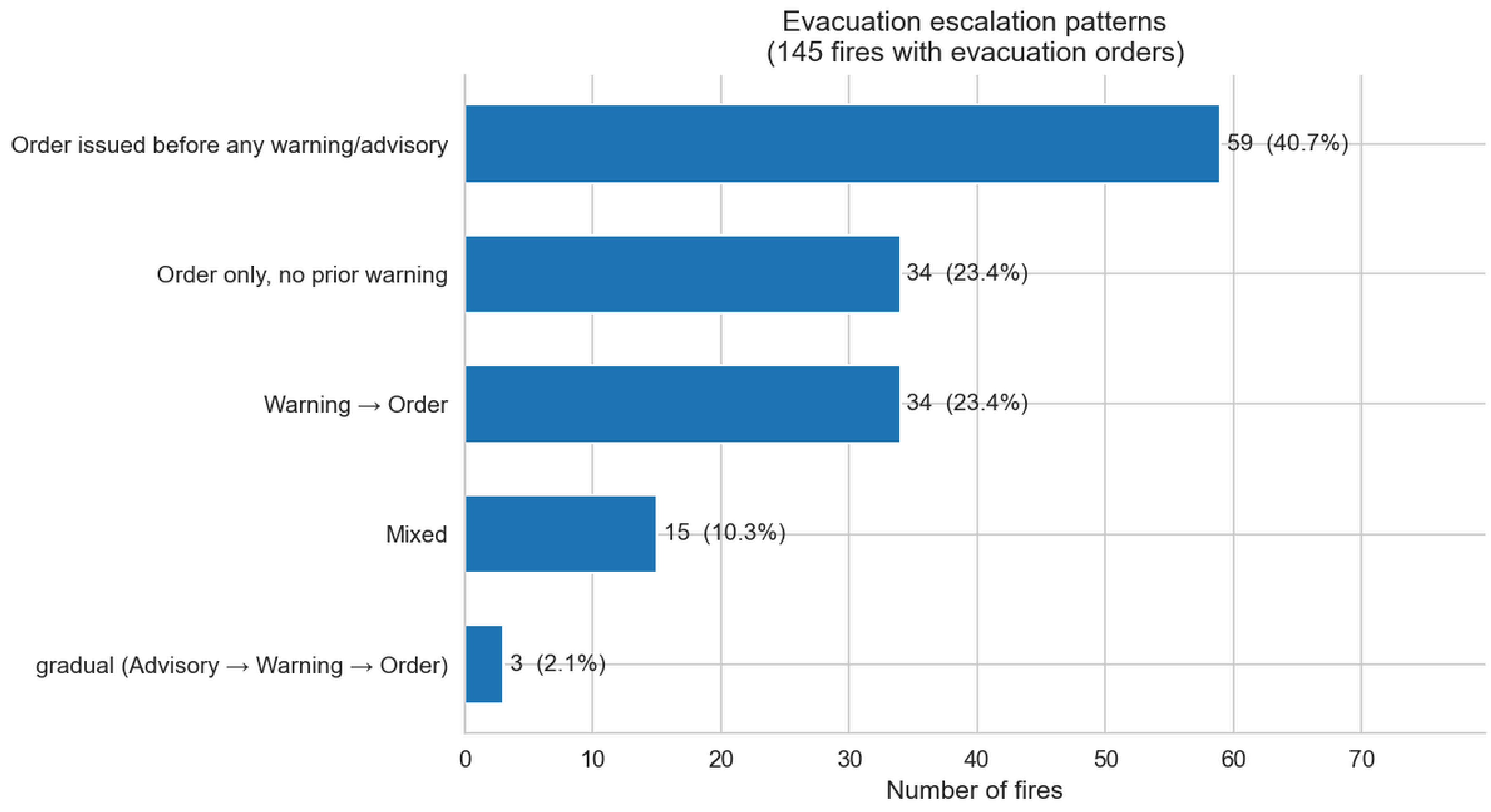



- ~15,000 CA wildfires
- ~11,600 monitored
- **ONLY ~250 with evacuation orders**

Only 1.8% of fires ever reach evacuation order,
which makes every minute of lag in those cases count

# Evacuation Order Pattern






**~ 2.1 %**

Advisory → Warning → Evacuation

**~ 64%**

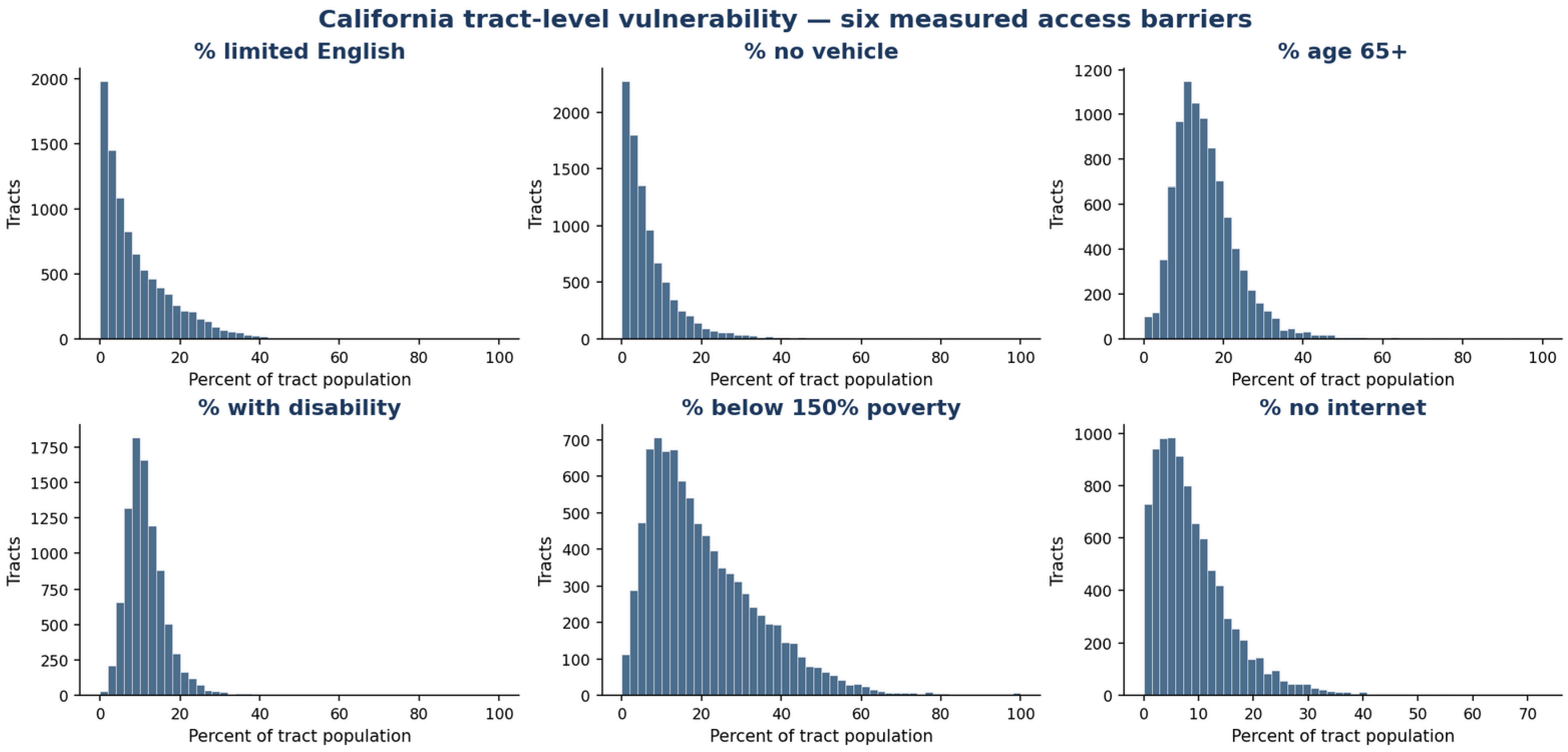
Straight to Evacuation without prior notice

# Vulnerability Distribution Across Tracts



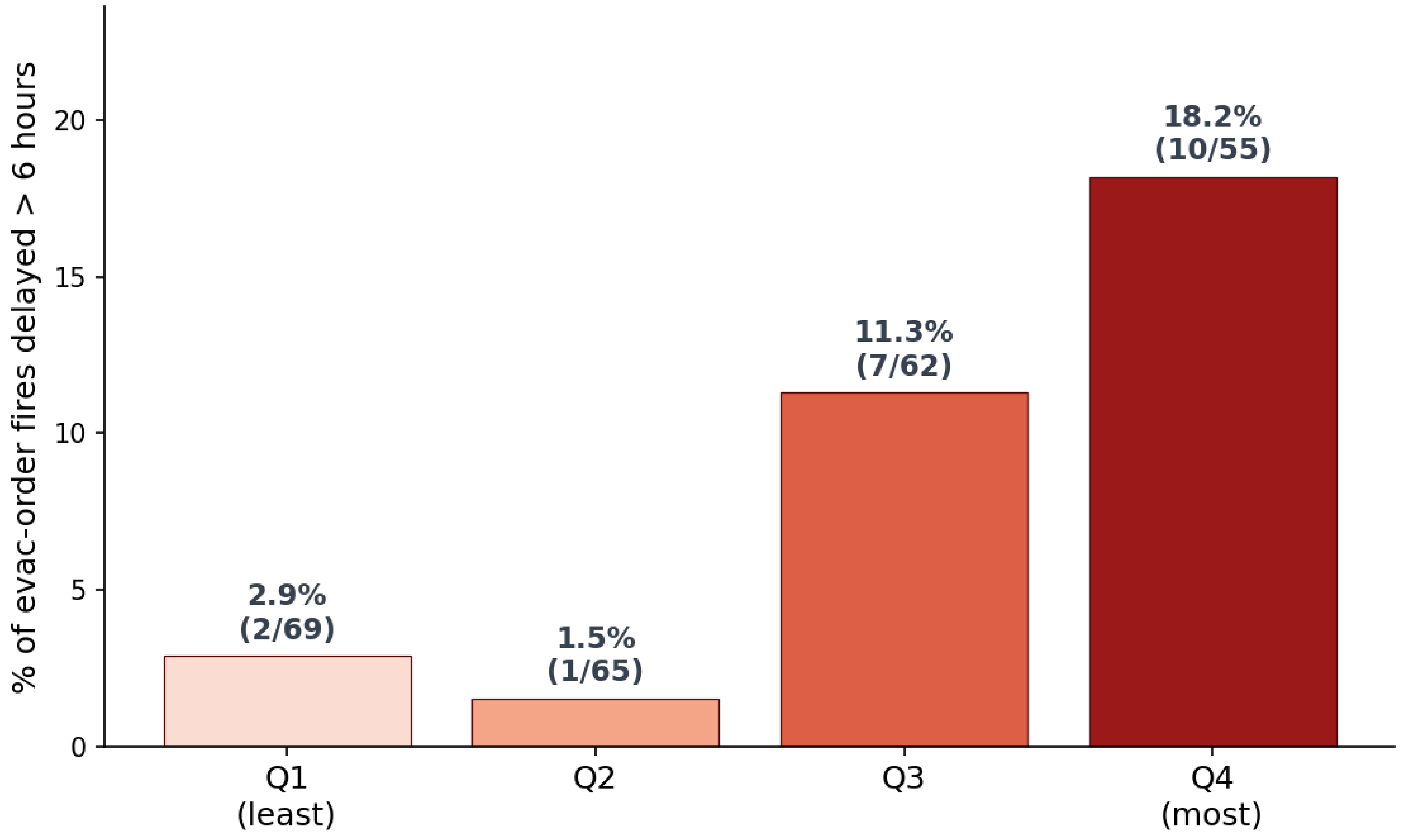
## Measured barriers

- Limited English
- No vehicle in the household
- Age 65+
- Disability
- Below 150% poverty
- No internet at home



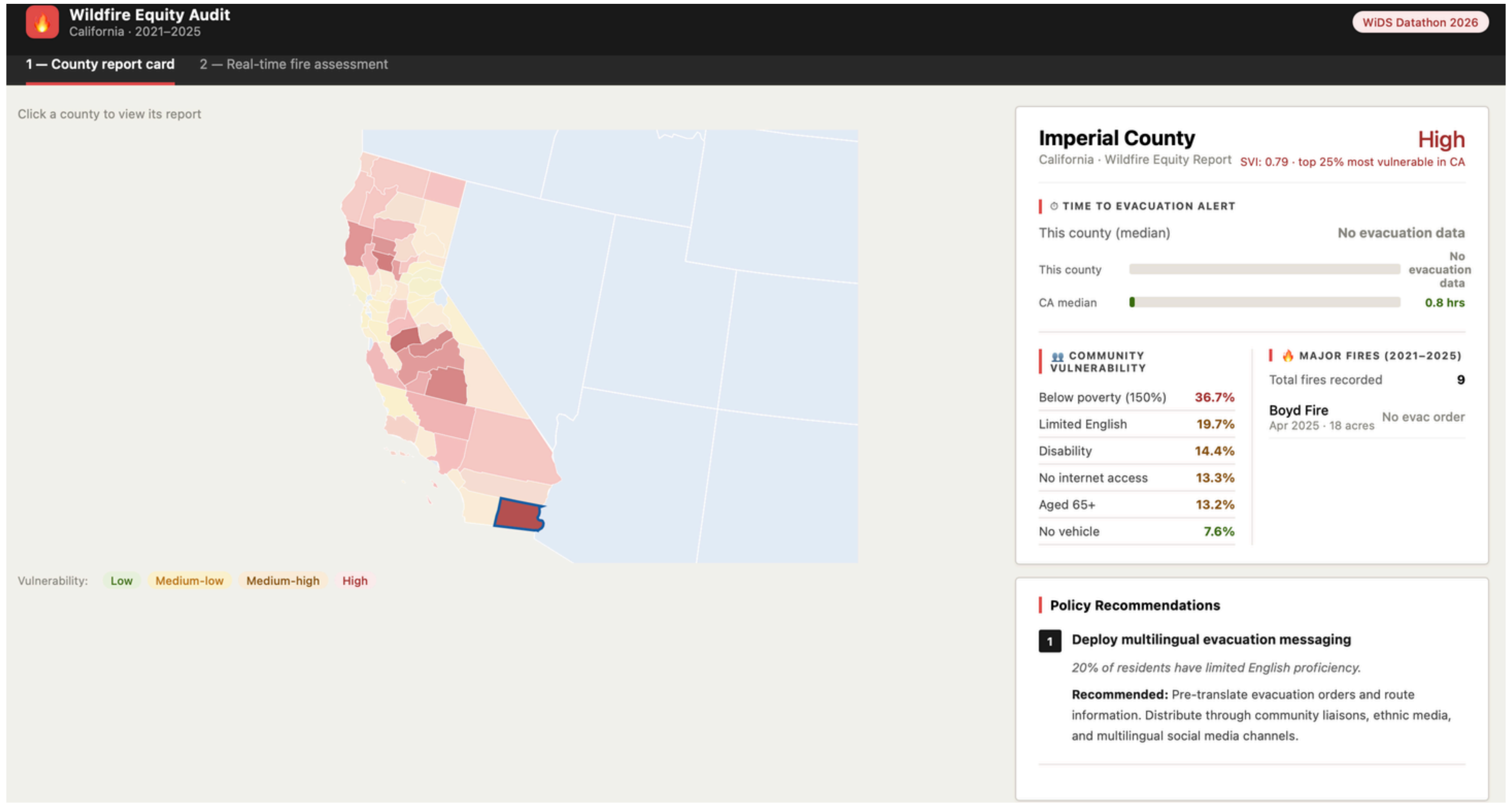
# Delay in Different Vulnerability






In the most vulnerable counties,
fires are 6× more likely to face delays over 6 hours.

# Our Solution:
# Wildfire Equity Dashboard for county emergency managers




### Tab 1: County Report Card

- Select any California county
- See alert lag, community vulnerability breakdown, and major fires history
- Get data-driven policy recommendations

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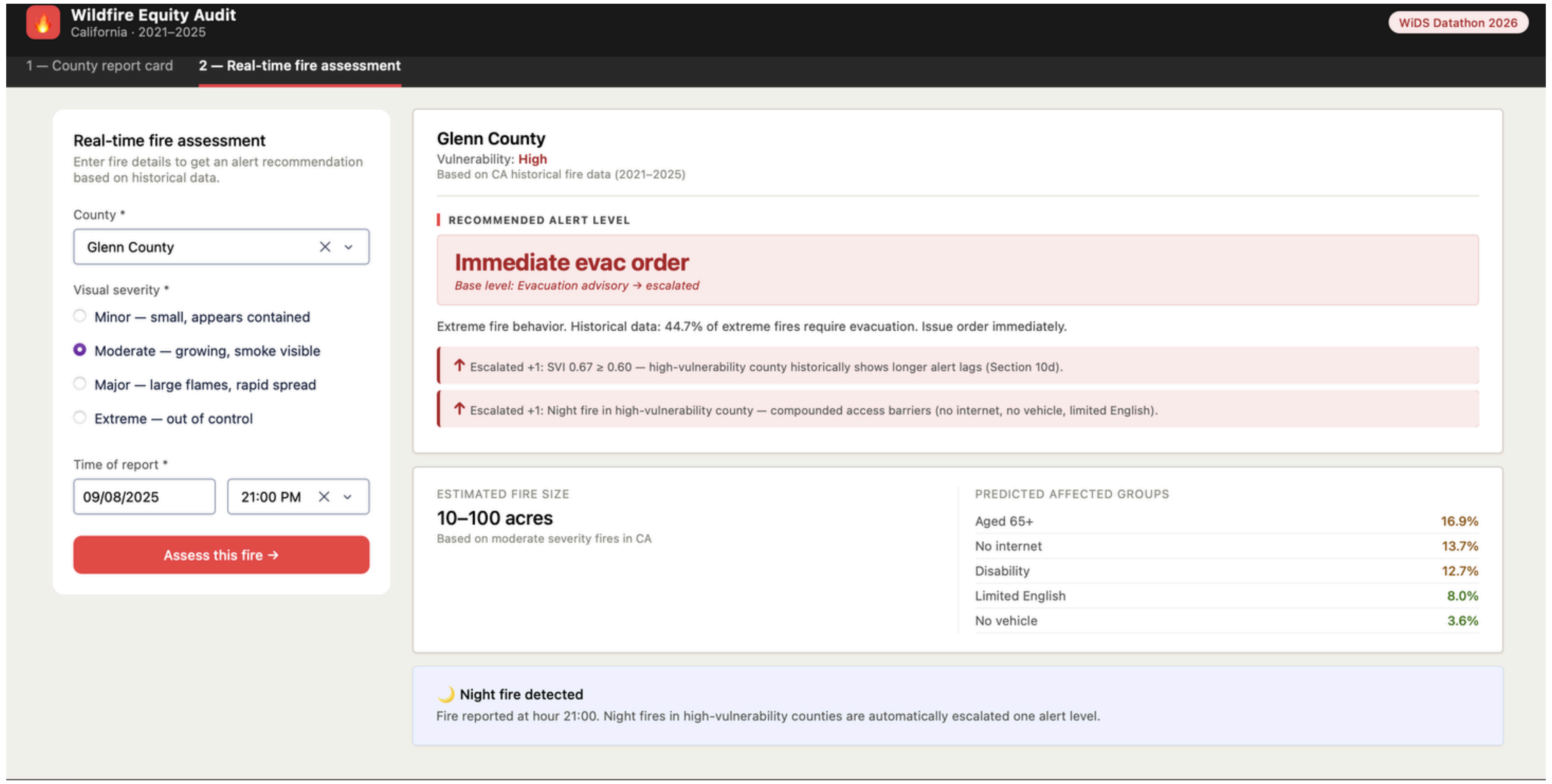


### Tab 2: Real-time Fire Assessment

- Input county, visual severity, and time
- Get a recommended alert level based on historical CA fire data
- Adjusts for community vulnerability and night fires

# Thank You

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